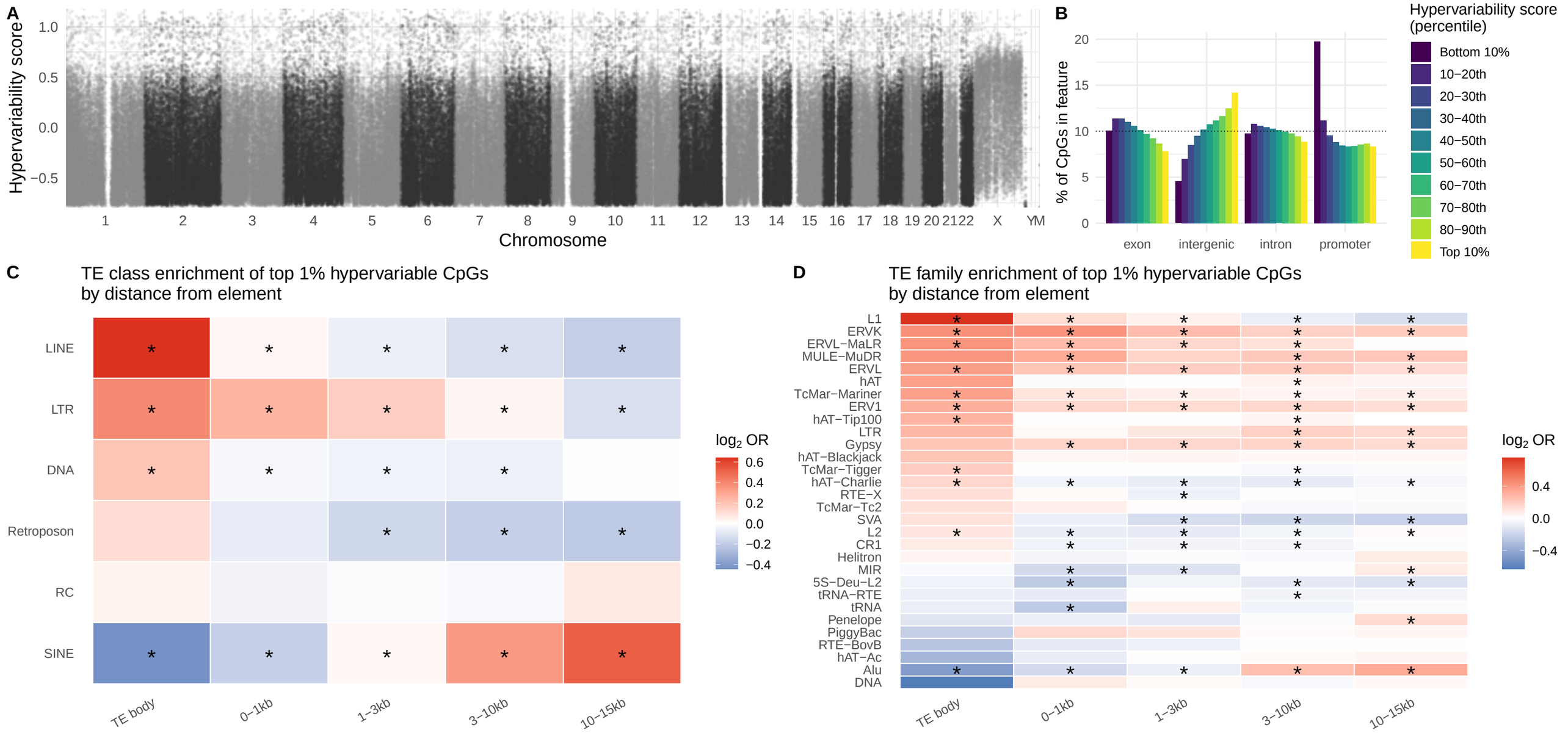

A
Hypervariability score
1.0
0.5
0.0
−0.5
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 X Y
Chromosome
B
% of CpGs in feature
20
15
10
5
0
exon
intergenic
intron
promoter
Hypervariability score (percentile)
Bottom 10%
10−20th
20−30th
30−40th
40−50th
50−60th
60−70th
70−80th
80−90th
Top 10%
C
TE class enrichment of top 1% hypervariable CpGs by distance from element
LINE
LTR
DNA
Retroposon
RC
SINE
TE body
0−1kb
1−3kb
3−10kb
10−15kb
log2 OR
0.6
0.4
0.2
0.0
−0.2
−0.4
D
TE family enrichment of top 1% hypervariable CpGs by distance from element
L1
ERVK
ERVL−MaLR
MULE−MuDR
ERVL
hAT
TcMar−Mariner
ERV1
hAT−Tip100
LTR
Gypsy
hAT−Blackjack
TcMar−Tigger
hAT−Charlie
RTE−X
TcMar−Tc2
SVA
L2
CR1
Helitron
MIR
5S−Deu−L2
tRNA−RTE
tRNA
Penelope
PiggyBac
RTE−BovB
hAT−Ac
Alu
DNA
TE body
0−1kb
1−3kb
3−10kb
10−15kb
log2 OR
0.4
0.0
−0.4
E
hvCpG enrichment around KZFP sites (KZFP as unit of replication)
Odds ratio (per−KZFP; pooled ± 95% CI)
1.00
0.10
0.01
KZFP peak
TE body
0−1kb
1−2kb
2−3kb
3−4kb
4−5kb
KZFP in TE
KZFP not in TE
F
hvCpG enrichment at KZFP−bound vs unbound elements
star = bound−vs−unbound difference (quasibinomial GLM; ns = not significant)
unbound
bound
All TEs
ERV1 + ERVK
Enrichment in top−1% hvCpG vs genome−wide; element−level, quasibinomial)
1.5
1.4
1.3
1.2
1.1
1.0
0.9
TE
TE+1kb
TE+2kb
TE+3kb